

Comparison to Critical Value:

- α is the user specified confidence level, e.g., $\alpha = 0.05$.
- Reject the null hypothesis if $T \geq \alpha$ or $T \geq (1 - \alpha)$.

Wilcoxon Signed Rank Test

The Wilcoxon signed rank test determines whether there is a difference in the average ranks of the two algorithms using paired samples. This test can also be described as determining if the median difference between paired observations is zero. The testing approach is as follows:

Hypothesis:

H_0 : Sample mean ranks are the same

H_1 : Sample mean ranks are different

Calculation process:

1. Let (A_i, B_i) be the paired performance results.
Let $D_i = A_i - B_i$ where $D_i > 0$ indicates algorithm A had better performance and $D_i < 0$ indicates algorithm B had better performance.
2. Sort the data based on the absolute values of differences $|D_1|, |D_2|, \dots, |D_n|$ in ascending order.
3. Assign a rank r_i to each observation. The smallest absolute value difference is assigned a rank of 1, the second smallest absolute value difference is assigned a rank of 2, $\dots$, and the largest absolute value difference is assigned a rank of n .
4. Assign a signed rank to each observation based on the rank and the original difference of the pair. That is:

$$S_i = \begin{cases} +r_i & \text{if } A_i - B_i > 0 \\ -r_i & \text{if } A_i - B_i < 0 \end{cases}$$

5. Let T_n be the sum of all ranks with a positive difference. This can be determined using an indicator function W_i defined as follows:

$$W_i = \begin{cases} 1 & \text{if } S_i > 0 \\ 0 & \text{if } S_i < 0 \end{cases}$$

$$T_n = \sum_{i=1}^n r_i \cdot W_i$$

Since the ranks r_i take on each value in the range $r_i = 1, 2, \dots, n$ (once and only once) T_n can also be written in terms of its observation as follows:

$$T_n = \sum_{i=1}^n i \cdot W_i$$